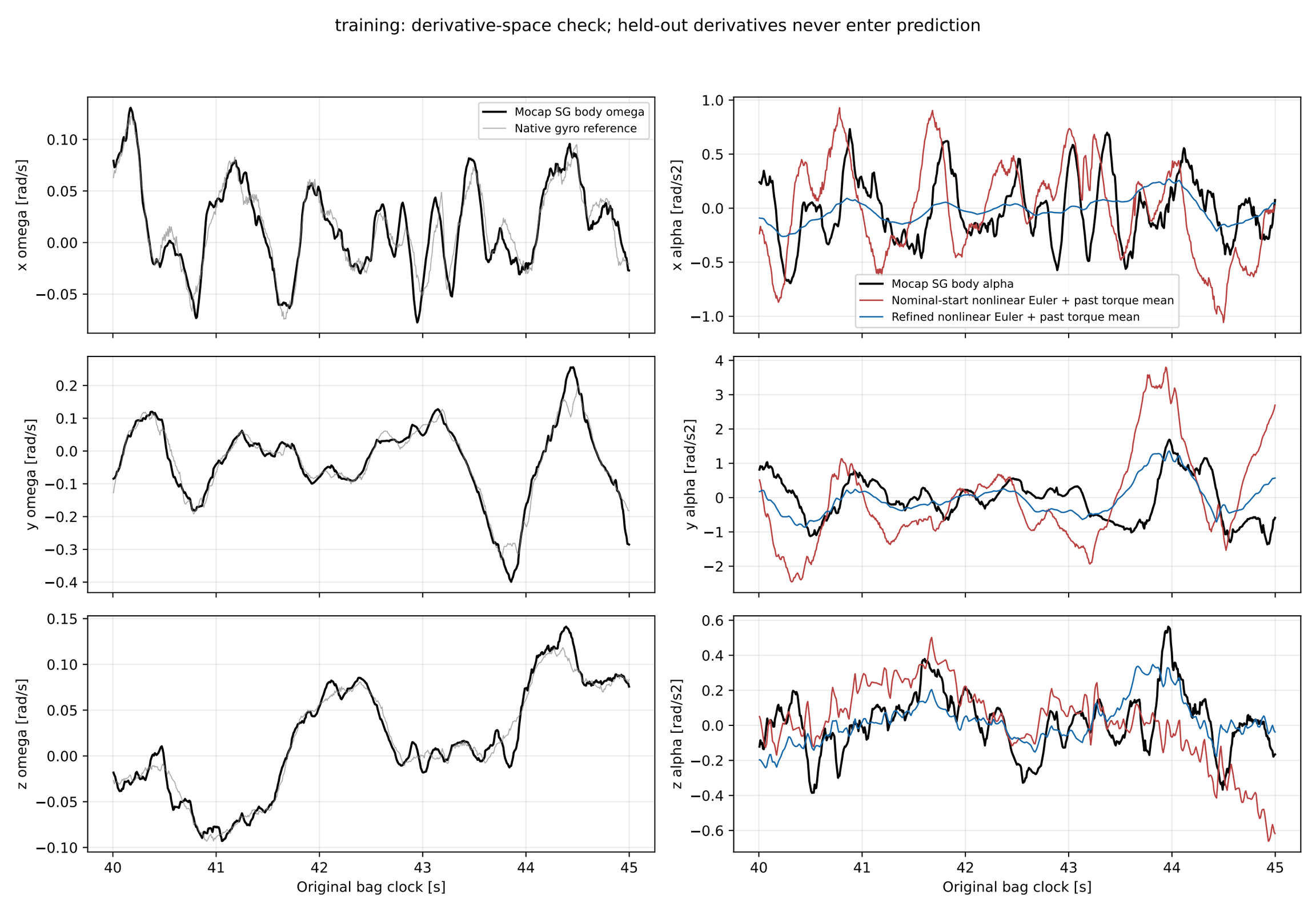
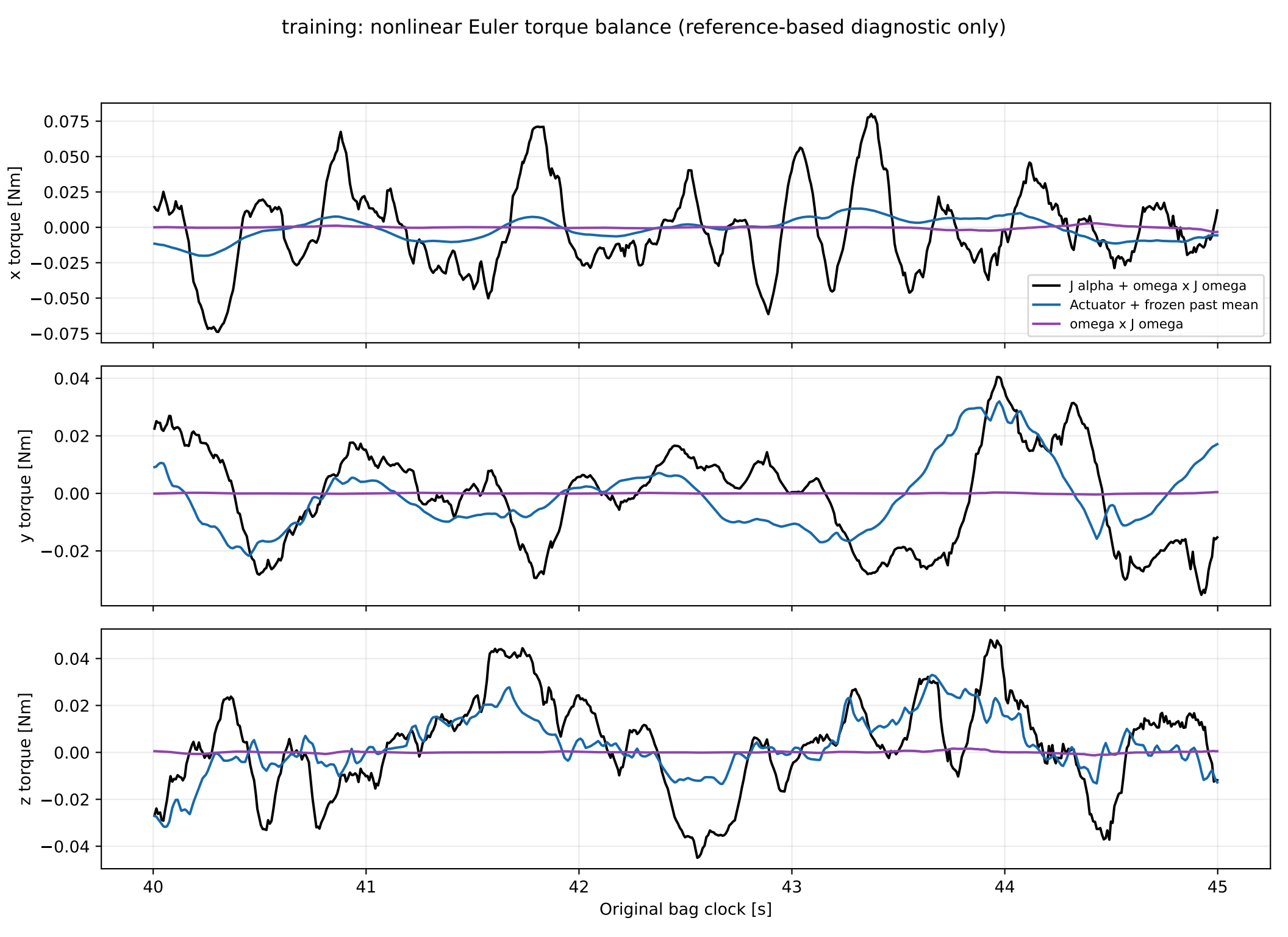
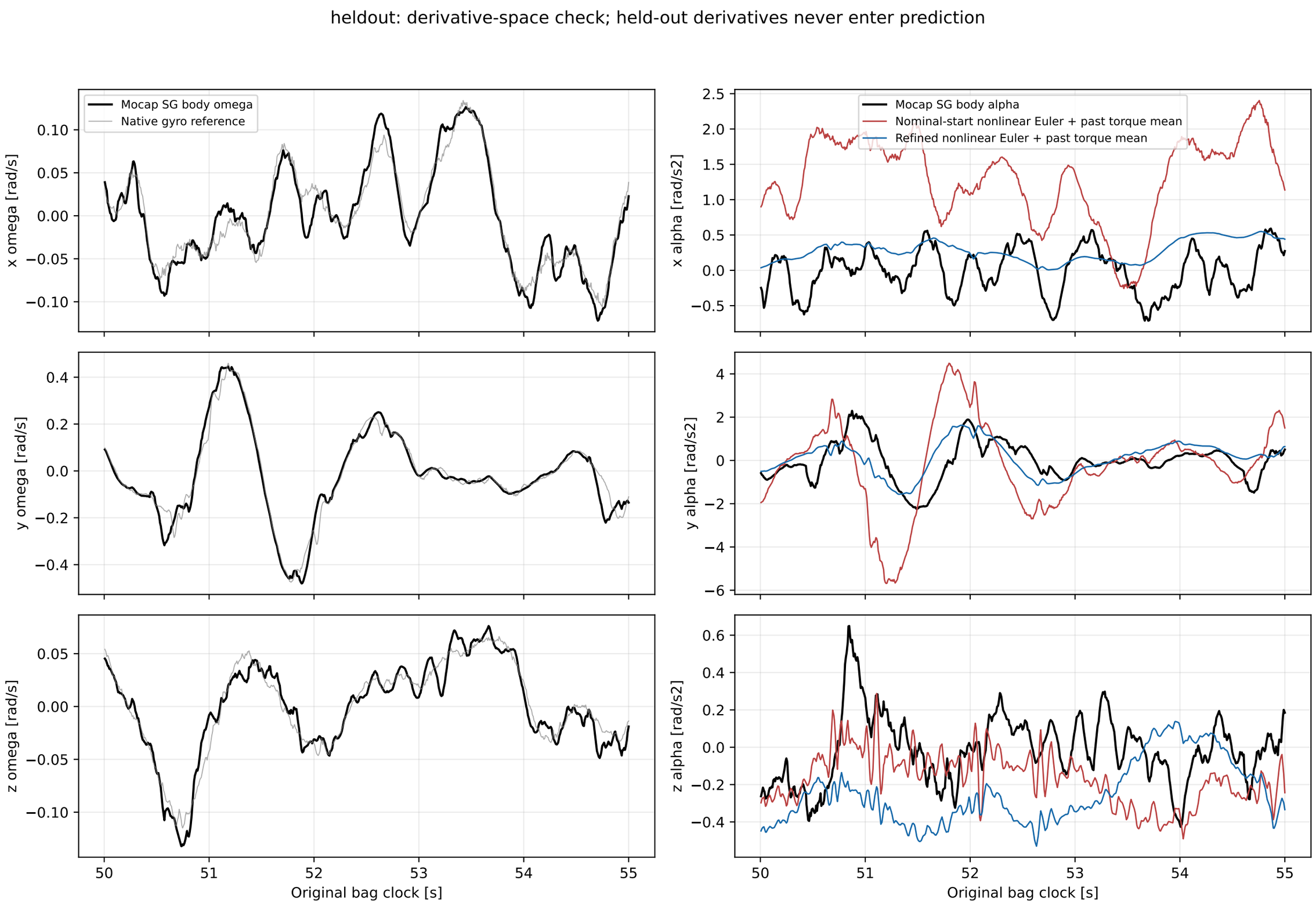
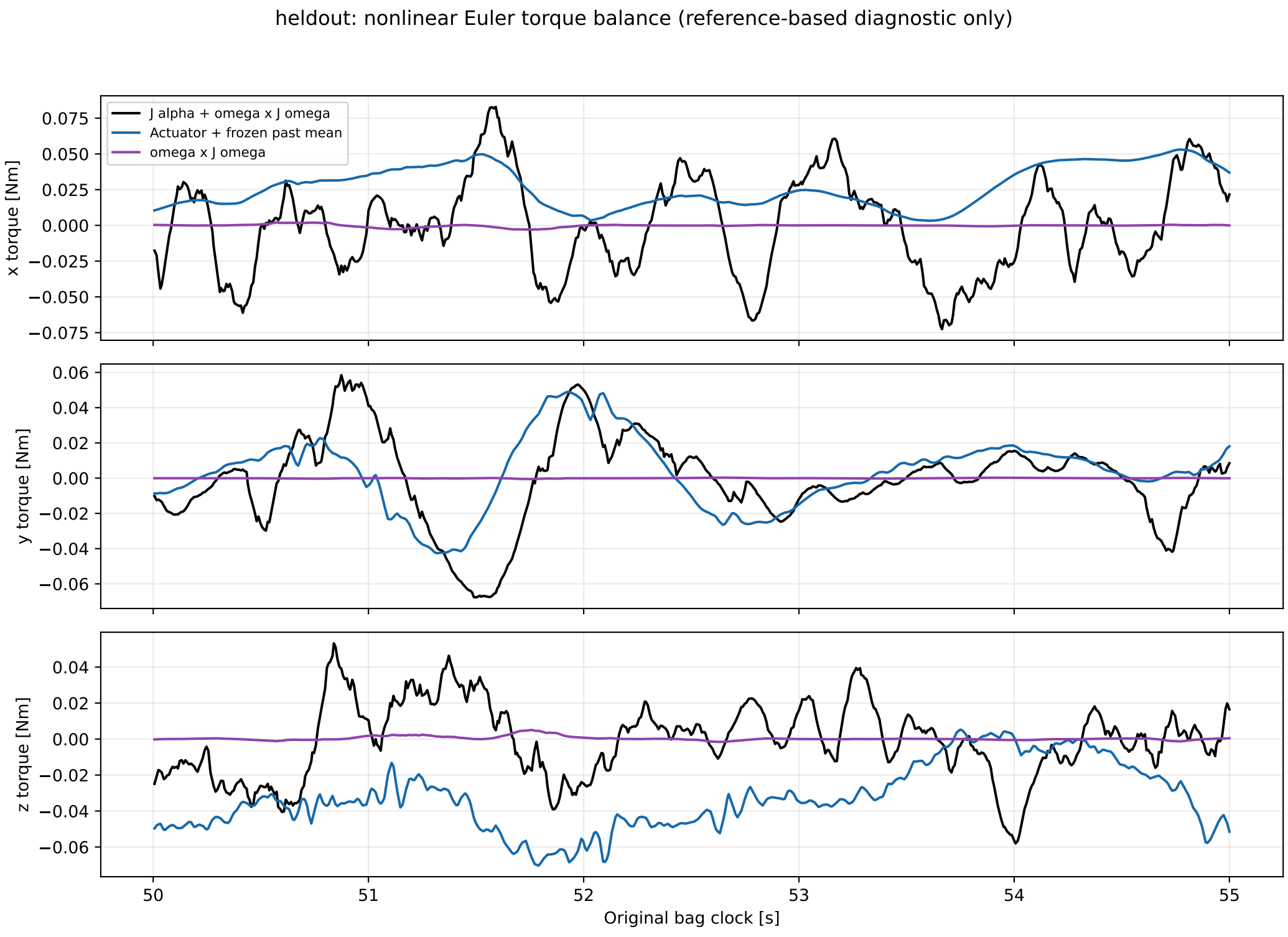

training: derivative-space check; held-out derivatives never enter prediction
Mocap SG body omega
Native gyro reference
x omega [rad/s]
y omega [rad/s]
z omega [rad/s]
Mocap SG body alpha
Nominal-start nonlinear Euler + past torque mean
Refined nonlinear Euler + past torque mean
x alpha [rad/s2]
y alpha [rad/s2]
z alpha [rad/s2]
Original bag clock [s]
Original bag clock [s]


training: nonlinear Euler torque balance (reference-based diagnostic only)
x torque [Nm]
y torque [Nm]
z torque [Nm]
J alpha + omega x J omega
Actuator + frozen past mean
omega x J omega
Original bag clock [s]
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
0.04
0.02
0.00
−0.02
−0.04
40
41
42
43
44
45


heldout: derivative-space check; held-out derivatives never enter prediction
Mocap SG body omega
Native gyro reference
Mocap SG body alpha
Nominal-start nonlinear Euler + past torque mean
Refined nonlinear Euler + past torque mean
x omega [rad/s]
y omega [rad/s]
z omega [rad/s]
x alpha [rad/s2]
y alpha [rad/s2]
z alpha [rad/s2]
Original bag clock [s]
Original bag clock [s]


heldout: nonlinear Euler torque balance (reference-based diagnostic only)
J alpha + omega x J omega
Actuator + frozen past mean
omega x J omega
x torque [Nm]
y torque [Nm]
z torque [Nm]
Original bag clock [s]
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
50
51
52
53
54
55